



American International University-Bangladesh (AIUB)

Department of Computer Science
Faculty of Science & Technology (FST)

Bangladesh Digital Health Registry (BDHR)

A Web Technologies Project

Supervised By

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Made By

Semester: Fall 25- 26		Section: F
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Product Requirements Document (PRD Table)

Phase	Item No.	Specification [Role]	Screen Definition	User Story	Acceptance Criteria
Feature: 1		Personal Health Management			
Phase 1	1	Patient Registration [Patient]	<Form> <input=Name> <input=Mobile Number> <input=Email> <input=Password> <button=Register>	As a new patient, I want to register to create my digital health account.	Mobile number and email must be unique. Password $\geq$ 8 characters. Verification OTP is sent. Redirect to login on success. Data securely stored in database.
Phase 1	2	Login with OTP [Patient]	<Form> <input=Email/Mobile> <input=OTP> <button=Login>	As a patient, I want to log in securely using OTP authentication .	OTP expires in 5 minutes. Three failed attempts lock account for 10 minutes. Redirect to dashboard on success.
Phase 1	3	View Medical History [Patient]	<List> Medical Reports, Prescriptions, Diagnoses, Vaccinations	As a patient, I want to view my entire medical history.	Sorted chronologically . Search by doctor/hospital. Download/print as PDF.

					Read-only access.
Phase 1	4	Book Appointment [Patient]	<Form> <dropdown=Doctor> <calendar=Date> <button=Confirm>	As a patient, I want to book a doctor's appointment online.	Doctor must be available. SMS/Email confirmation sent. Synced to both dashboards.
Phase 1	5	Grant Access to Doctor [Patient]	<Modal> <select=Doctor> <button=Grant Access>	As a patient, I want to grant temporary access to my records to a doctor.	Access valid for defined duration (24 h). Patient notified when records viewed. Consent logged for audit.
Phase 1	6	Privacy Settings [Patient]	<Toggle List> <option=Show/Hide Sensitive Data>	As a patient, I want to control visibility of my mental or sensitive health data.	Visibility editable anytime. Doctor needs patient consent. All changes timestamped.
Phase 1	7	NFC Health Card [Patient]	<NFC Scan> <button=Authenticate>	As a patient, I want to use my NFC card for quick hospital check-ins.	Authenticates using secure token. Fetches profile instantly. Logs location and time of use.
Feature: 2		Digital Clinical Practice			

Phase 1	8	Doctor Registration & Verification [Doctor]	<Form> <input=Name> <input=License No> <upload=License Proof> <button=Submit>	As a doctor, I want to register and verify my credentials.	License validated by Admin. Approved doctors gain access. Rejected doctors notified via email.
Phase 1	9	Doctor Dashboard [Doctor]	<Dashboard> Widgets: Appointments, Reports, Patient Search	As a doctor, I want an overview of appointments and patients.	Shows upcoming appointments. Search patients by ID/name. Refresh every 24 h.
Phase 1	10	Update Diagnosis & Prescription [Doctor]	<Form> <textarea=Diagnosis> <textarea=Prescription> > <button=Save>	As a doctor, I want to record diagnosis and prescriptions digitally.	Auto-sign with digital signature. Linked to patient record. Patient notified immediately.
Phase 1	11	Access Lab Reports [Doctor]	<List> <Reports with Upload Date> <button=View>	As a doctor, I want to access uploaded lab reports.	Filter by test type or date. Request additional tests. Logs every access.
Phase 1	12	Emergency Access [Doctor]	<Modal> <input=Reason> <button=Request Access>	As a doctor, I want to access critical data in emergencies with consent.	Requires reason for access. Auto-report sent to DGHS. Access revoked after session.

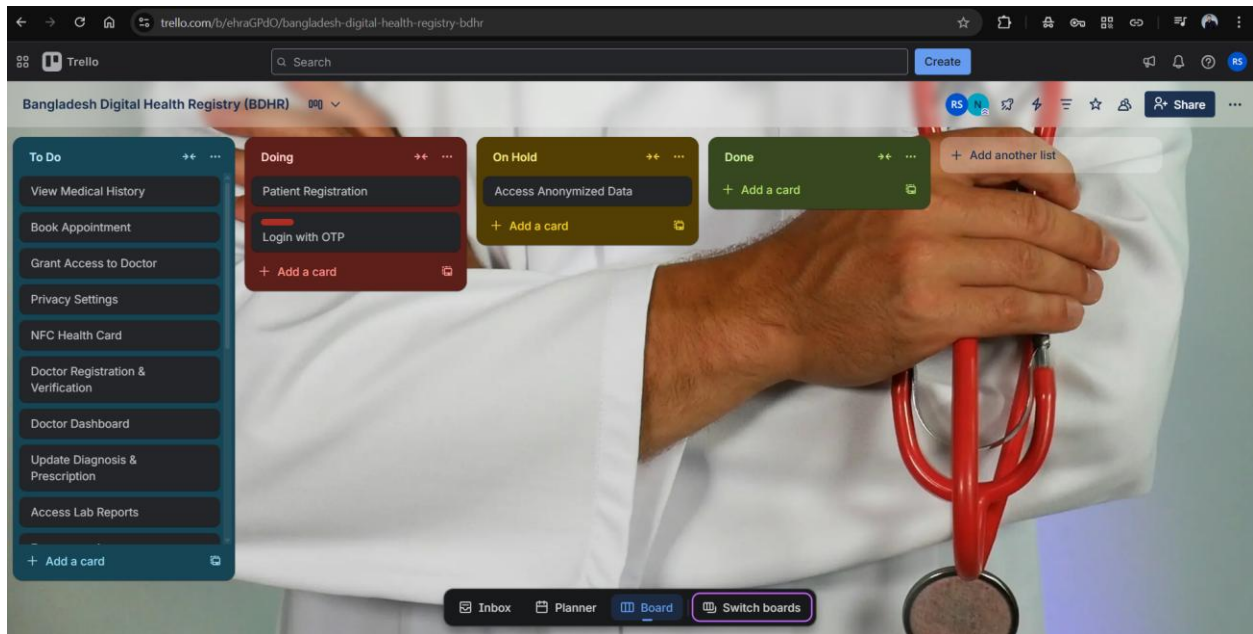
Feature: 3		Administration & Facility Management			
Phase 1	13	Doctor Verification [Admin]	<Table> Columns: Name, License No, Status, Action	As an admin, I want to verify doctor licenses.	Cross-checked via DGHS database. Approve/Reject buttons available. Changes logged for audit.
Phase 1	14	Manage Admissions [Admin]	<Form> <input=Patient ID> <dropdown=Ward> <button=Admit>	As an admin, I want to record patient admissions.	Admission time and date recorded. Automatically updates patient profile. Discharge option available.
Phase 1	15	Upload Test Reports [Admin/Lab]	<Upload Form> <input=Patient ID> <upload=PDF/Image>	As a lab staff, I want to upload test reports for patients.	Valid file types only (PDF/JPG). Stored under correct patient ID. Timestamp and uploader recorded.

Phase 1	16	Data Monitoring & Audit [Admin]	<Dashboard> Graphs: Records Updated, Errors Flagged	As an admin, I want to monitor data quality and validity.	Detect incomplete records. Audit logs searchable by date/user. Alerts for unusual activity.
Feature: 4		Digital Prescription Mangement			
Phase 1	17	Login [Pharmacist]	<Form> <input=Email> <input=Password> <button=Login>	As a pharmacist, I want to log into the system securely.	Two-factor authentication enabled. Redirect to prescription dashboard on login.
Phase 1	18	Verify Digital Prescription [Pharmacist]	<Modal> <input=Prescription ID> <button=Verify>	As a pharmacist, I want to verify doctor's digital signature on a prescription.	Signature validated via certificate. Invalid prescriptions flagged. Log stored for each verification.
Phase 1	19	Dispense Medication [Pharmacist]	<Form> <input=Patient ID> <input=Drug Name> <button=Dispense>	As a pharmacist, I want to record dispensed medicines.	Only verified prescriptions can be dispensed. Record includes drug batch and expiry. Patient notified

					of dispense status.
Feature: 5		Government Access			
Phase 1	20	Access Anonymized Data [DGHS]	<Dashboard> Filters: Region, Disease Type, Date Range	As a DGHS official, I want to analyze national health data without personal identifiers.	Only aggregated data visible. Export to CSV/Excel available. Access restricted by role.
Phase 1	21	Monitor System Compliance [DGHS]	<Report View> <Graph=Data Compliance Score>	As a DGHS auditor, I want to ensure that all institutions follow data standards.	Weekly compliance score generated. Auto-alert for low scoring institutions.
Phase 1	22	Generate Public Health Reports [DGHS]	<Dashboard> Charts: Disease Trend, Immunization Coverage	As a DGHS officer, I want to generate reports for policy planning.	Reports auto-generated from aggregated data. Exportable to PDF and Excel. Scheduled monthly updates.
Phase 1	23	Oversee Hospital Registrations [DGHS]	<Table> Columns: Hospital Name, District, Status, Action	As a DGHS administrator, I want to approve and monitor registered hospitals.	Only licensed hospitals approved. Status change notifications sent to admins.

					Audit trail maintained.
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Trello Board



Link:

<https://trello.com/invite/b/690df3eff6152712c5edea60/ATTI8ab9a0420114066984800df3adf27daeA3FA6607/bangladesh-digital-health-registry-bdhr>